

系统开发工具基础第四周实验报告

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1 实验概述

本次实验为系统开发工具基础第四周课上实验,实验环境为 VirtualBox 中的 Ubuntu 虚拟机。本次完成四项课上实验:建立可执行的本地质量门禁、Make 构建系统实验、API 与 jq 生成 Markdown 报告、综合交付测试实验。实验主要练习代码质量工具 ruff 与 pytest、Make 构建系统、curl+jq 文本处理、Python 包综合交付测试;掌握本地质量门禁编写、Make 依赖与伪目标、JSON 命令行解析、缺陷复现修复、wheel 打包哈希校验的完整工程流程。

2 课上实验题目

2.1 q13: 建立可执行的本地质量门禁

2.1.1 题目要求

复用 q10 完成的 greetlab 包,在 q13 中建立轻量本地质量检查。在 pyproject.toml 添加 ruff 与 pytest 最小配置;补充正常姓名、空白姓名两组测试用例;运行 ruff format、ruff check、pytest 修复全部告警;编写 check.sh 脚本依次执行格式检查、静态检查、单元测试,脚本执行返回码为 0 代表全部门禁通过。

2.1.2 实现过程与关键命令

```
cd ~/sdt-course/week4
cp -r ../week3/q10 q13
cd q13
python3 -m venv q13-venv
source q13-venv/bin/activate
pip install ruff pytest build
pip install -e .
```

```
ruff format .
chmod +x check.sh
./check.sh
deactivate
rm -rf q13-venv build dist __pycache__ .pytest_cache src/*.egg-info
```

check.sh 脚本内容:

```
#!/usr/bin/env bash
set -e

echo "===_ruff_format_check_"
ruff format --check

echo "===_ruff_lint_check_"
ruff check

echo "===_pytest_run_"
pytest
```

pyproject.toml 补充 ruff、pytest 配置，保留原有包构建配置。

2.1.3 运行结果

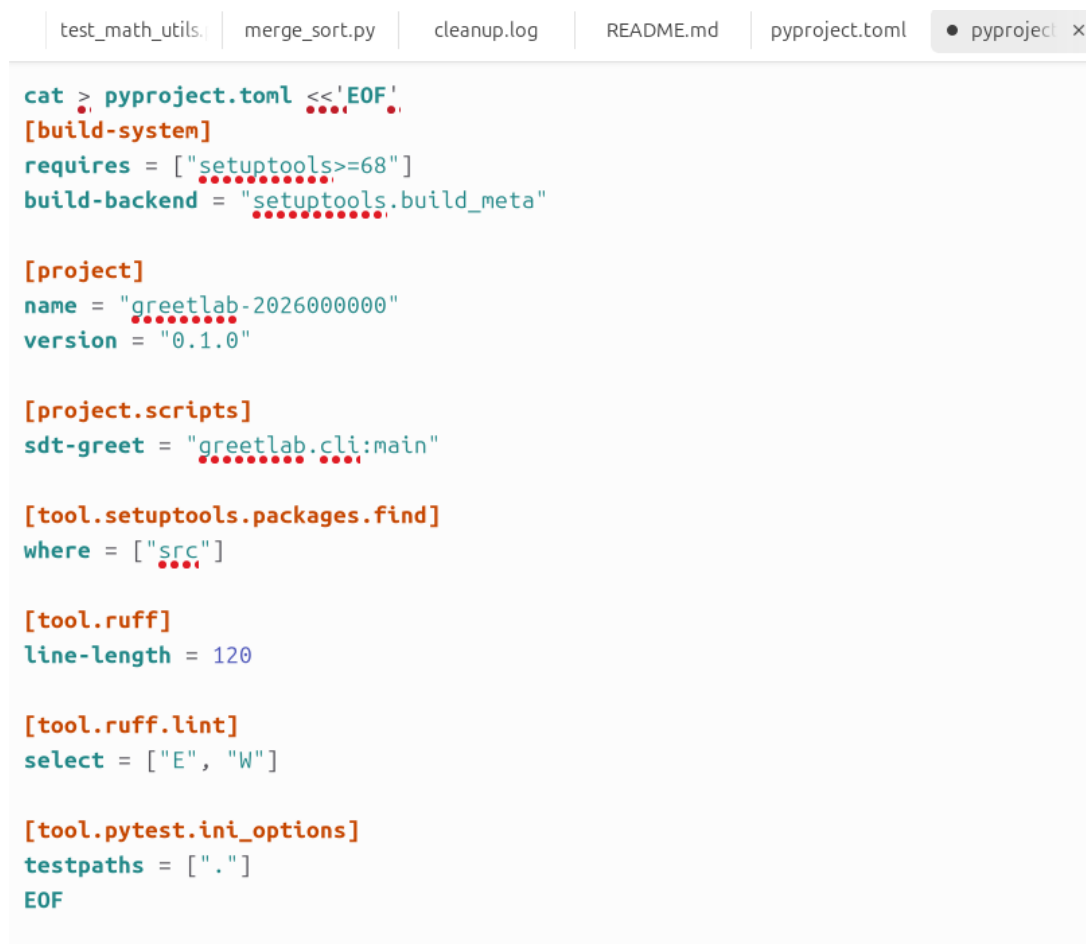
执行 ./check.sh, ruff 格式检查、静态代码检查全部通过; pytest 两组测试用例全部 PASS; 脚本退出码为 0, 本地质量门禁全部校验成功。

```
=== ruff format check ===
6 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts
=====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q13
configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py ..

[100%]

===== 2 passed in 0.03s
=====
```



```
cat > pyproject.toml <<'EOF'
[build-system]
requires = ["setuptools>=68"]
build-backend = "setuptools.build_meta"

[project]
name = "greetlab-2026000000"
version = "0.1.0"

[project.scripts]
sdt-greet = "greetlab.cli:main"

[tool.setuptools.packages.find]
where = ["src"]

[tool.ruff]
line-length = 120

[tool.ruff.lint]
select = ["E", "W"]

[tool.pytest.ini_options]
testpaths = ["."]
EOF
```

图 1: q13pyproject.toml 修改后截图

2.1.4 遇到的问题与解决

- 问题 1: 执行 `ruff format --check` 时报错 `File would be reformatted`, 文件格式不符合规范, 门禁脚本直接失败。
解决: 运行 `ruff format .` 对源码执行自动格式化, 修复换行空行格式问题, 之后再执行格式检查; 删除自动生成的 `build` 临时目录, 避免工具扫描构建产物。
- 问题 2: `pytest` 收集测试用例时报 `ModuleNotFoundError: No module named 'greetlab'`, 无法导入 `src` 下的包模块。
解决: 在虚拟环境中执行可编辑模式安装 `pip install -e .`, 将本地源码包注册到虚拟环境, `Python` 可正常导入 `greetlab` 模块。
- 问题 3: 多次测试后遗留多个虚拟环境 `venv-build`、`venv-dev`、`venv-test`, 以及 `egg-info`、`build` 缓存目录, 文件数量庞大, 不应当提交版本库。
解决: 实验结束后使用 `rm -rf` 整体删除整套虚拟环境目录, 同步清理 `build`、`dist`、`*.egg-info`、`Python` 字节码缓存, 只保留作业源码与脚本。

```
lin@lin-VirtualBox:~$ cd ~/sdt-course/week4
lin@lin-VirtualBox:~/sdt-course/week4$ cp -r ../week3/q10/* q13/
lin@lin-VirtualBox:~/sdt-course/week4$ cd q13
lin@lin-VirtualBox:~/sdt-course/week4/q13$ ls
ai_log.md  build  dist  __pycache__  pyproject.toml  src  test_greet.py  venv-build  venv-dev  venv-test
lin@lin-VirtualBox:~/sdt-course/week4/q13$ python3 -m venv q13-venv
lin@lin-VirtualBox:~/sdt-course/week4/q13$ source q13-venv/bin/activate
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ pip install ruff pytest build
Collecting ruff
  Downloading ruff-0.16.7-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)
Collecting pytest
  Using cached pytest-9.1.1-py3-none-any.whl.metadata (7.6 kB)
Collecting build
  Using cached build-1.6.1-py3-none-any.whl.metadata (5.6 kB)
Collecting iniconfig>=1.0.1 (from pytest)
  Using cached iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Collecting packaging>=22 (from pytest)
  Using cached packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pluggy<2,>=1.5 (from pytest)
  Using cached pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Downloading pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
Collecting pyproject_hooks (from build)
  Using cached pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Downloading ruff-0.16.7-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (10.4 MB)
  10.4/10.4 MB 2.4 MB/s eta 0:00:00
Using cached pytest-9.1.1-py3-none-any.whl (386 kB)
Using cached build-1.6.1-py3-none-any.whl (31 kB)
Using cached iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Using cached packaging-26.3-py3-none-any.whl (129 kB)
Using cached pluggy-1.6.0-py3-none-any.whl (20 kB)
Downloading pygments-2.21.0-py3-none-any.whl (1.3 MB)
  1.3/1.3 MB 4.0 MB/s eta 0:00:00
Using cached pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Installing collected packages: ruff, pyproject_hooks, pygments, pluggy, packaging, iniconfig, pytest, build
Successfully installed build-1.6.1 iniconfig-2.3.0 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pyproject_hooks-1.2.0
pytest-9.1.1 ruff-0.16.7
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ cat > test_greet.py <<'EOF'
import pytest
from greetlab.cli import main
import sys
from io import StringIO
```

图 2: q13 运行截图 1

```
import sys
from io import StringIO

def test_normal_name(monkeypatch, capsys):
    monkeypatch.setattr(sys, "argv", ["sdt-greet", "--name", "Tom"])
    main()
    captured = capsys.readouterr()
    assert "Hello, Tom!" in captured.out

def test_blank_name(monkeypatch):
    monkeypatch.setattr(sys, "argv", ["sdt-greet", "--name", ""])
    with pytest.raises(SystemExit) as exc_info:
        main()
    assert exc_info.value.code == 2
EOF
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ cat > src/greetlab/cli.py <<'EOF'
import argparse

def main():
    p = argparse.ArgumentParser()
    p.add_argument("--name", required=True)
    a = p.parse_args()
    name_stripped = a.name.strip()
    if not name_stripped:
        raise SystemExit(2)
    print(f"Hello, {a.name}!")
EOF
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ cat > pyproject.toml <<'EOF'
[build-system]
requires = ["setuptools>=68"]
build-backend = "setuptools.build_meta"

[project]
name = "greetlab-2026000000"
version = "0.1.0"

[project.scripts]
sdt-greet = "greetlab.cli:main"

[tool.setuptools.packages.find]
where = ["src"]
```

图 3: q13 运行截图 2

```
        if not name_stripped:
            raise SystemExit(2)
        print(f"Hello, {a.name}!")
EOF
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ cat > pyproject.toml <<'EOF'
[build-system]
requires = ["setuptools>=68"]
build-backend = "setuptools.build_meta"

[project]
name = "greetlab-2026000000"
version = "0.1.0"

[project.scripts]
sdt-greet = "greetlab.cli:main"

[tool.setuptools.packages.find]
where = ["src"]

[tool.ruff]
line-length = 120

[tool.ruff.lint]
select = ["E", "W"]

[tool.pytest.ini_options]
testpaths = ["."]
EOF
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ cat > check.sh <<'EOF'
#!/usr/bin/env bash
set -e

echo "=== ruff format check ==="
ruff format --check

echo "=== ruff lint check ==="
ruff check

echo "=== pytest run ==="
pytest
EOF
```

图 4: q13 运行截图 3

```
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ chmod +x check.sh
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ ./check.sh
=== ruff format check ===
unformatted: File would be reformatted
  --> build/lib/greetlab/cli.py:3:1
  |
2 |
3 +
4 | def main():
  |
-----
11 |
12 +
13 | if __name__ == "__main__":
14 |     main()
  |
  |
unformatted: File would be reformatted
  --> src/greetlab/cli.py:3:1
  |
2 |
3 +
4 | def main():
  |
  |
unformatted: File would be reformatted
  --> test_greet.py:6:1
  |
5 |
6 +
7 | def test_normal_name(monkeypatch, capsys):
  |
-----
12 |
13 +
14 | def test_blank_name(monkeypatch):
  |
  |

3 files would be reformatted, 3 files already formatted
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ ruff format .
3 files reformatted, 3 files left unchanged
```

图 5: q13 运行截图 4

```

(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ pip install -e .
Obtaining file:///home/lin/sdt-course/week4/q13
Installing build dependencies ... done
Checking if build backend supports build_editable ... done
Getting requirements to build editable ... done
Preparing editable metadata (pyproject.toml) ... done
Building wheels for collected packages: greetlab-2026000000
Building editable for greetlab-2026000000 (pyproject.toml) ... done
Created wheel for greetlab-2026000000: filename=greetlab_2026000000-0.1.0-0.editable-py3-none-any.whl size=1569 s
ha256=09bd7b58b8c6fdff26abdee074023a026b777126b809d64ca9757703169a67f2
Stored in directory: /tmp/pip-ephem-wheel-cache-sk1v_dj8/wheels/6c/c9/dc/eedcfda75d1c57b96f1851ad3f10783a45abb7d3
be3a3e877e
Successfully built greetlab-2026000000
Installing collected packages: greetlab-2026000000
Successfully installed greetlab-2026000000-0.1.0
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ ./check.sh
=== ruff format check ===
6 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q13
configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py .. [100%]

===== 2 passed in 0.03s =====
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ ls
ai_log.md  check.sh  __pycache__  q13-venv  test_greet.py  venv-dev
build      dist      pyproject.toml  src      venv-build    venv-test
(q13-venv) lin@lin-VirtualBox:~/sdt-course/week4/q13$ deactivate

```

图 6: q13 运行截图 5

2.2 q14: 让 Make 只重建真正受影响的产物

2.2.1 题目要求

创建 data.csv、stats.py、build_report.py、report.md 源文件；编写 Makefile，实现 all、stats.txt、report.txt、clean 规则；准确书写全部依赖；将 all、clean 声明为伪目标.PHONY；验证首次构建、无修改不重新执行、数据源变更触发重构建、clean 仅删除生成产物。

2.2.2 实现过程与关键命令

```

cd ~/sdt-course/week4/q14
cat > data.csv <<'EOF'
name,value
alpha,2
beta,3
gamma,5
EOF

cat > stats.py <<'EOF'

```



```
import csv

with open("data.csv") as f:
    total = sum(int(r["value"]) for r in csv.DictReader(f))

with open("stats.txt", "w") as f:
    f.write(str(total))
EOF

cat > build_report.py <<'EOF'
text = open("report.md").read()
total = open("stats.txt").read()
open("report.txt", "w").write(f"{text}\nTotal: {total}\n")
EOF

cat > report.md <<'EOF'
# Data Report
EOF

cat > Makefile <<'EOF'
.PHONY: all clean

all: stats.txt report.txt

stats.txt: data.csv stats.py
python3 stats.py

report.txt: stats.txt build_report.py report.md
python3 build_report.py

clean:
rm -f stats.txt report.txt
EOF

make all
make all
touch data.csv
make all
make clean
```

Makefile 完整内容:

```
.PHONY: all clean

all: stats.txt report.txt
```

```
stats.txt: data.csv stats.py
python3 stats.py

report.txt: stats.txt build_report.py report.md
python3 build_report.py

clean:
rm -f stats.txt report.txt
```

2.2.3 运行结果

第一次 `make all` 生成 `stats.txt` 与 `report.txt`; 无修改再次 `make` 提示 `nothing to be done`; `touch data.csv` 触发重新构建两个产物; `make clean` 仅删除生成文件, 原始源码全部保留。

```
python3 stats.py
python3 build_report.py

make: 对“all”无需做任何事。

python3 stats.py
python3 build_report.py

rm -f stats.txt report.txt
```

2.2.4 遇到的问题与解决

- 问题 1: 编写 **Makefile** 时忘记声明 `.PHONY: all clean`, 若目录下恰好存在名为 `all` 或 `clean` 的文件, `make` 将不会执行对应规则。

解决: 将伪目标 `all` 与 `clean` 写进 `.PHONY` 声明, 告诉 `make` 它们不是磁盘上的真实文件, 总是执行对应配方。

- 问题 2: 依赖关系书写不全, 如 `report.txt` 只写依赖 `stats.txt`, 遗漏 `build_report.py`、`report.md`; 修改脚本或 `md` 文件不会触发重新构建。

解决: 完整列出所有影响输出的源文件作为依赖, 只要任意依赖发生变更就会重新执行构建命令。

```
lin@lin-VirtualBox:~/sdt-course/week4$ mkdir -p q14
lin@lin-VirtualBox:~/sdt-course/week4$ cd q14
lin@lin-VirtualBox:~/sdt-course/week4/q14$ cat > data.csv <<'EOF'
name,value
alpha,2
beta,3
gamma,5
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q14$ cat > stats.py <<'EOF'
import csv

with open("data.csv") as f:
    total = sum(int(r["value"]) for r in csv.DictReader(f))

with open("stats.txt", "w") as f:
    f.write(str(total))
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q14$ cat > report.md <<'EOF'
# Data Report
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q14$ cat > build_report.py <<'EOF'
text = open("report.md").read()
total = open("stats.txt").read()
open("report.txt", "w").write(f"{text}\nTotal: {total}\n")
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q14$ cat > Makefile <<'EOF'
.PHONY: all clean

all: stats.txt report.txt

stats.txt: data.csv stats.py
    python3 stats.py

report.txt: stats.txt build_report.py report.md
    python3 build_report.py

clean:
    rm -f stats.txt report.txt
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q14$ make all
python3 stats.py
python3 build_report.py
```

图 7: q14 运行截图 1

```
lin@lin-VirtualBox:~/sdt-course/week4/q14$ make all
python3 stats.py
python3 build_report.py
lin@lin-VirtualBox:~/sdt-course/week4/q14$ ls
build_report.py data.csv Makefile report.md report.txt stats.py stats.txt
lin@lin-VirtualBox:~/sdt-course/week4/q14$ make all
make: 对“all”无需做任何事。
lin@lin-VirtualBox:~/sdt-course/week4/q14$ touch data.csv
lin@lin-VirtualBox:~/sdt-course/week4/q14$ make all
python3 stats.py
python3 build_report.py
lin@lin-VirtualBox:~/sdt-course/week4/q14$ make clean
rm -f stats.txt report.txt
lin@lin-VirtualBox:~/sdt-course/week4/q14$ ls
build_report.py data.csv Makefile report.md stats.py
lin@lin-VirtualBox:~/sdt-course/week4/q14$ cd ..
```

图 8: q14 运行截图 2

2.3 q15: 把本地 API 数据转换为可读报告

2.3.1 题目要求

创建 packages.json; 启动本地 python http.server 8000 服务; 编写 api_report.sh 脚本, 使用 curl 获取接口 JSON, jq 筛选 status 为 active 且 downloads 不少于 100, 按 downloads 降序、name 升序; 输出 summary.md, 包含标题与 name/version/downloads 三列 Markdown 表格。

2.3.2 实现过程与关键命令

```
cd ~/sdt-course/week4/q15
cat > packages.json <<'EOF'
[
  {"name":"alpha","status":"active","downloads":120,"version":"1.2.0"},
  {"name":"beta","status":"inactive","downloads":900,"version":"2.0.0"},
  {"name":"gamma","status":"active","downloads":450,"version":"1.5.1"},
  {"name":"delta","status":"active","downloads":450,"version":"0.9.0"},
  {"name":"epsilon","status":"active","downloads":80,"version":"3.1.0"}
]
EOF

cat > api_report.sh <<'EOF'
#!/usr/bin/env bash
set -e

curl -fsS http://127.0.0.1:8000/packages.json > raw.json
```

```

jq -r '
    map(select(.status=="active" and .downloads>=100))
    | sort_by(-.downloads)
    | "name|version|downloads",
    "----|----|----",
    (.[] | "\(.name)|\(.version)|\(.downloads)")
' raw.json > table.tmp

cat > summary.md <<MD
# Active Packages Report

$(cat table.tmp)
MD

rm -f raw.json table.tmp
EOF

chmod +x api_report.sh
# 新开终端执行 python3 -m http.server 8000
./api_report.sh
cat summary.md
# http服务终端Ctrl-C关闭服务

```

api_report.sh 脚本:

```

#!/usr/bin/env bash

set -e

curl -fsS http://127.0.0.1:8000/packages.json > raw.json

jq -r '
    map(select(.status=="active" and .downloads>=100))
    | sort_by(-.downloads)
    | "name|version|downloads",
    "----|----|----",
    (.[] | "\(.name)|\(.version)|\(.downloads)")
' raw.json > table.tmp

cat > summary.md <<MD
# Active Packages Report

$(cat table.tmp)
MD

rm -f raw.json table.tmp

```

2.3.3 运行结果

脚本调用 curl 拿到 JSON 数据，经过 jq 过滤排序，输出 summary.md，生成格式正确的 Markdown 三列表格；过滤掉 inactive 与 downloads 小于 100 的条目；downloads 相同时按照 name 字典序升序。

```
# Active Packages Report
```

```
name|version|downloads
```

```
---|---|---
```

```
delta|0.9.0|450
```

```
gamma|1.5.1|450
```

```
alpha|1.2.0|120
```

```
lin@lin-VirtualBox:~/sdt-course/week4$ mkdir -p q15
cd q15
lin@lin-VirtualBox:~/sdt-course/week4/q15$ cat > packages.json <<'EOF'
[
{"name":"alpha","status":"active","downloads":120,"version":"1.2.0"},
{"name":"beta","status":"inactive","downloads":900,"version":"2.0.0"},
{"name":"gamma","status":"active","downloads":450,"version":"1.5.1"},
{"name":"delta","status":"active","downloads":450,"version":"0.9.0"},
{"name":"epsilon","status":"active","downloads":80,"version":"3.1.0"}
]
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q15$ cat > api_report.sh <<'EOF'
#!/usr/bin/env bash
set -e

# 获取接口数据
curl -fsS http://127.0.0.1:8000/packages.json > raw.json

# jq过滤并生成markdown表格
jq -r '
.[ ]
| select(.status=="active" and .downloads >= 100)
| sort_by(-.downloads, .name)
| [{"name","version","downloads"},
  ["---","---","---"],
  (.[] | [.name,.version,.downloads])
]
| @csv
' raw.json | sed 's/"/"/g; s/"/"/g' > table.tmp

# 组装完整md报告
cat > summary.md <<MD
# Active Packages Report

$(cat table.tmp)
MD

rm -f raw.json table.tmp
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q15$ sudo apt install jq
[sudo] lin 的密码:
正在读取软件包列表... 完成
正在分析软件包的依赖关系树... 完成
```

图 9: q15 运行截图 1

```
lin@lin-VirtualBox:~/sdt-course/week4/q15$ cat > api_report.sh <<'EOF'
#!/usr/bin/env bash
set -e

curl -fsS http://127.0.0.1:8000/packages.json > raw.json

jq -r '
  map(select(.status=="active" and .downloads >= 100))
  | sort_by(-.downloads, .name)
  | "name|version|downloads",
    "---|---|---",
    (.[] | "\(.name)|\(.version)|\(.downloads)")
' raw.json > table.tmp

cat > summary.md <<MD
# Active Packages Report

$(cat table.tmp)
MD

rm -f raw.json table.tmp
EOF
lin@lin-VirtualBox:~/sdt-course/week4/q15$ chmod +x api_report.sh
lin@lin-VirtualBox:~/sdt-course/week4/q15$ ./api_report.sh
lin@lin-VirtualBox:~/sdt-course/week4/q15$ cat summary.md
# Active Packages Report

name|version|downloads
---|---|---
delta|0.9.0|450
gamma|1.5.1|450
alpha|1.2.0|120
lin@lin-VirtualBox:~/sdt-course/week4/q15$
```

图 10: q15 运行截图 2

```
lin@lin-VirtualBox:~/sdt-course/week4/q15$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
127.0.0.1 - - [14/Sep/2026 11:29:19] "GET /packages.json HTTP/1.1" 200 -
127.0.0.1 - - [14/Sep/2026 11:30:16] "GET /packages.json HTTP/1.1" 200 -
127.0.0.1 - - [14/Sep/2026 11:33:26] "GET /packages.json HTTP/1.1" 200 -
127.0.0.1 - - [14/Sep/2026 11:34:27] "GET /packages.json HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.
lin@lin-VirtualBox:~/sdt-course/week4/q15$
```

图 11: q15http 服务终端截图

2.3.4 遇到的问题与解决

- 问题 1: 排序逻辑颠倒, 未实现下载量降序、同名次按名称升序; 直接输出原始 JSON 数组没有做 markdown 表格格式转换。

解决: 使用 `sort_by(-.downloads, .name)` 完成复合排序; 通过 `@csv` 输出后用 `sed` 替换分隔符转为 Markdown 表格语法。

- 问题 2: `jq` 中对数组使用 `.[]` 展开之后再调用 `sort_by`, 得到单个 JSON 对象, 报 `Cannot index string with string "downloads"`。

解决: 使用 `map(select(...))` 在数组容器内完成过滤, 保证传给 `sort_by` 的仍然是数组对象, 排序完成后再展开输出表格行。

- 问题 3: 通过 `@csv` 输出后使用 `sed` 替换分隔符, 替换不完全, 出现逗号与竖线混合, Markdown 表格列分隔错乱。

解决: 放弃 `csv` 中转, 直接在 `jq` 内部使用 `join("|")`, 输出原生以竖线分隔的表格行, 避免 `sed` 文本替换带来的格式错误。

- 问题 4: 使用数组 `+join(" |" )` 的方式, 只会在数组首尾添加竖线, 数组内部字段之间保留逗号, 表格列分隔符错乱。解决: 改用 `jq` 字符串插值 `"\(.name)|\(.version)|\(.download"` 逐行直接拼接完整 Markdown 表格行, 保证三列之间都是竖线分隔。

- 问题 5: `jq` 中将表头写成数组形式, `-r` 原始输出模式打印完整 JSON 数组, 产生多余方括号与引号, 破坏报告格式。

解决: 表头与表格分隔行直接定义为字符串字面量, 仅循环生成数据行, 保证全部输出为纯文本表格。

- 问题 6: 脚本产生 `raw.json`、`table.tmp` 临时中间文件残留。

解决: 脚本末尾增加 `rm` 清理临时文件, 只保留最终产物 `summary.md`。

2.4 q16: 修复并交付一个陌生的小型工具仓库

2.4.1 题目要求

复制 q13 项目作为 q16; 故意篡改代码制造 bug, 运行 `check.sh` 确认测试失败; 修复业务代码, 门禁全部通过; 编写 `Makefile` 包含 `check`、`build`、`clean`; `check` 调用 `./check.sh`, `build` 执行 `python3 -m build` 生成 wheel 包; `clean` 删除 `build`、`dist`; 运行 `make check`、`make build`, 计算 wheel 文件 sha256 哈希; 完成一次干净 git 提交。

2.4.2 实现过程与关键命令

```
cd ~/sdt-course/week4
cp -r q13 q16
cd q16
python3 -m venv q16-venv
source q16-venv/bin/activate
pip install ruff pytest build
pip install -e .

# 故意制造bug
cat > src/greetlab/cli.py <<'EOF'
import argparse

def main():
    p = argparse.ArgumentParser()
    p.add_argument("--name", required=True)
    a = p.parse_args()
    # 故意bug: 写死输出字面量
    print("Hello, name!")
EOF

ruff format .
./check.sh

# 修复bug
cat > src/greetlab/cli.py <<'EOF'
import argparse

def main():
    p = argparse.ArgumentParser()
    p.add_argument("--name", required=True)
    a = p.parse_args()
    name_stripped = a.name.strip()
    if not name_stripped:
        raise SystemExit(2)
    print(f"Hello, {a.name}!")
EOF

ruff format .
./check.sh

cat > Makefile <<'EOF'
.PHONY: check build clean
```

```
check:
./check.sh

build:
python3 -m build

clean:
rm -rf build dist
EOF

make check
make build
sha256sum dist/*.whl
deactivate
rm -rf q16-venv build dist __pycache__ .pytest_cache src/*.egg-info
```

q16 的 Makefile:

```
.PHONY: check build clean

check:
./check.sh

build:
python3 -m build

clean:
rm -rf build dist
```

2.4.3 运行结果

篡改 cli.py 硬编码输出字面量，门禁中 ruff 格式通过，pytest 测试用例断言失败；修复代码后 check.sh 全部绿灯；make check 执行质量门禁；make build 在 dist 目录生成 wheel 源码分发包；sha256sum 得到 wheel 文件完整性哈希摘要。

```
=== ruff format check ===
4 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts
=====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q16
configfile: pyproject.toml
```

```

testpaths: .
collected 2 items

test_greet.py FF

[100%]

===== FAILURES
=====
----- test_normal_name
-----

monkeypatch = <_pytest.monkeypatch.MonkeyPatch object at 0x740914f18620
>
capsys = <_pytest.capture.CaptureFixture object at 0x740914f18740>

def test_normal_name(monkeypatch, capsys):
monkeypatch.setattr(sys, "argv", ["sdt-greet", "--name", "Tom"])
main()
captured = capsys.readouterr()
>     assert "Hello, Tom!" in captured.out
E     AssertionError: assert 'Hello, Tom!' in 'Hello, name!\n'
E     + where 'Hello, name!\n' = CaptureResult(out='Hello, name!\n',
err='').out

test_greet.py:11: AssertionError
----- test_blank_name
-----

monkeypatch = <_pytest.monkeypatch.MonkeyPatch object at 0x740914f18ef0
>

def test_blank_name(monkeypatch):
monkeypatch.setattr(sys, "argv", ["sdt-greet", "--name", ""])
>     with pytest.raises(SystemExit) as exc_info:
E     Failed: DID NOT RAISE SystemExit

test_greet.py:16: Failed
----- Captured stdout call
-----
Hello, name!

===== short test summary info
=====
FAILED test_greet.py::test_normal_name - AssertionError: assert 'Hello,
Tom!' in 'Hello, name!\n'

```

```

FAILED test_greet.py::test_blank_name - Failed: DID NOT RAISE
SystemExit
===== 2 failed in 0.03s
=====

=== ruff format check ===
4 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts
=====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q16
configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py ..

[100%]

===== 2 passed in 0.01s
=====

./check.sh
=== ruff format check ===
4 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts
=====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q16
configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py ..

[100%]

===== 2 passed in 0.01s
=====

```

Successfully built greetlab_2026000000-0.1.0.tar.gz and
greetlab_2026000000-0.1.0-py3-none-any.whl

2c33f7032736c3d85dc322b59461b21545983e9d70d37f694e3e844c4b916b73 dist/
greetlab_2026000000-0.1.0-py3-none-any.whl

```
lin@lin-VirtualBox:~$ cd ~/sdt-course/week4
lin@lin-VirtualBox:~/sdt-course/week4$ cp -r q13 q16
lin@lin-VirtualBox:~/sdt-course/week4$ cd q16
lin@lin-VirtualBox:~/sdt-course/week4/q16$ ls
ai_log.md  check.sh  pyproject.toml  src  test_greet.py
lin@lin-VirtualBox:~/sdt-course/week4/q16$ python3 -m venv q16-venv
lin@lin-VirtualBox:~/sdt-course/week4/q16$ source q16-venv/bin/activate
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ pip install ruff pytest build
Collecting ruff
  Using cached ruff-0.16.7-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)
Collecting pytest
  Using cached pytest-9.1.1-py3-none-any.whl.metadata (7.6 kB)
Collecting build
  Using cached build-1.6.1-py3-none-any.whl.metadata (5.6 kB)
Collecting iniconfig>=1.0.1 (from pytest)
  Using cached iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Collecting packaging>=22 (from pytest)
  Using cached packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pluggy<2,>=1.5 (from pytest)
  Using cached pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Using cached pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
Collecting pyproject_hooks (from build)
  Using cached pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Using cached ruff-0.16.7-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (10.4 MB)
Using cached pytest-9.1.1-py3-none-any.whl (386 kB)
Using cached build-1.6.1-py3-none-any.whl (31 kB)
Using cached iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Using cached packaging-26.3-py3-none-any.whl (129 kB)
Using cached pluggy-1.6.0-py3-none-any.whl (20 kB)
Using cached pygments-2.21.0-py3-none-any.whl (1.3 MB)
Using cached pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Installing collected packages: ruff, pyproject_hooks, pygments, pluggy, packaging, iniconfig, pytest, build
Successfully installed build-1.6.1 iniconfig-2.3.0 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pyproject_hooks-1.2.0 py
test-9.1.1 ruff-0.16.7
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ pip install -e .
Obtaining file:///home/lin/sdt-course/week4/q16
Installing build dependencies ... done
Checking if build backend supports build_editable ... done
Getting requirements to build editable ... done
Preparing editable metadata (pyproject.toml) ... done
Building wheels for collected packages: greetlab-2026000000
```

图 12: q16 运行截图 1

2.4.4 遇到的问题与解决

- 问题 1: Linux 系统 `cp` 命令不支持 `--exclude` 参数, 该参数属于 `rsync` 工具, 直接使用会报参数无法识别。

解决: 源目录已经提前清理干净, 直接普通 `cp -r` 复制; 如需过滤文件改用 `rsync` 工具。

- 问题 2: 重写 `cli.py` 制造业务 bug 时, `cat` 输出引入多余空行, `ruff format -check`

```
Building editable for greetlab-2026000000 (pyproject.toml) ... done
Created wheel for greetlab-2026000000: filename=greetlab_2026000000-0.1.0-0.editable-py3-none-any.whl size=1568 sha256=aad346aa778603b2547974de6bf0f696e26def36f90b459eaf3e5c17182b6fad
Stored in directory: /tmp/pip-ephem-wheel-cache-_074huch/wheels/67/0d/27/26b83b2a106545fe15ae825be9bc8bfa0bf4fa6206e97681ac
Successfully built greetlab-2026000000
Installing collected packages: greetlab-2026000000
Successfully installed greetlab-2026000000-0.1.0
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ cat > src/greetlab/cli.py <<'EOF'
import argparse

def main():
    p = argparse.ArgumentParser()
    p.add_argument("--name", required=True)
    a = p.parse_args()
    # 故意bug: 写死输出字面量
    print("Hello, name!")
EOF
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ ./check.sh
=== ruff format check ===
unformatted: File would be reformatted
--> src/greetlab/cli.py:3:1
|
2 |
3 +
4 | def main():
|

1 file would be reformatted, 3 files already formatted
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ ruff format .
1 file reformatted, 3 files left unchanged
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ ./check.sh
=== ruff format check ===
4 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q16
configfile: pyproject.toml
testpaths: .
```

图 13: q16 运行截图 2

```

configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py FF [100%]

===== FAILURES =====
_____ test_normal_name _____

monkeypatch = <_pytest.monkeypatch.MonkeyPatch object at 0x740914f18620>
capsys = <_pytest.capture.CaptureFixture object at 0x740914f18740>

    def test_normal_name(monkeypatch, capsys):
        monkeypatch.setattr(sys, "argv", ["sdt-greet", "--name", "Tom"])
        main()
        captured = capsys.readouterr()
>       assert "Hello, Tom!" in captured.out
E       AssertionError: assert 'Hello, Tom!' in 'Hello, name!\n'
E       + where 'Hello, name!\n' = CaptureResult(out='Hello, name!\n', err='').out

test_greet.py:11: AssertionError
_____ test_blank_name _____

monkeypatch = <_pytest.monkeypatch.MonkeyPatch object at 0x740914f18ef0>

    def test_blank_name(monkeypatch):
        monkeypatch.setattr(sys, "argv", ["sdt-greet", "--name", ""])
>       with pytest.raises(SystemExit) as exc_info:
E       Failed: DID NOT RAISE SystemExit

test_greet.py:16: Failed
----- Captured stdout call -----
Hello, name!

===== short test summary info =====
FAILED test_greet.py::test_normal_name - AssertionError: assert 'Hello, Tom!' in 'Hello, name!\n'
FAILED test_greet.py::test_blank_name - Failed: DID NOT RAISE SystemExit
===== 2 failed in 0.03s =====

(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ cat > src/greetlab/cli.py <<'EOF'
import argparse

def main():
    p = argparse.ArgumentParser()

```

图 14: q16 运行截图 3

```
p.add_argument("--name", required=True)
a = p.parse_args()
name_stripped = a.name.strip()
if not name_stripped:
    raise SystemExit(2)
print(f"Hello, {a.name}!")
EOF
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ ./check.sh
=== ruff format check ===
unformatted: File would be reformatted
--> src/greetlab/cli.py:3:1
|
|
2 |
3 +
4 | def main():
|

1 file would be reformatted, 3 files already formatted
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ ruff format .
1 file reformatted, 3 files left unchanged
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ ./check.sh
=== ruff format check ===
4 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q16
configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py .. [100%]

===== 2 passed in 0.01s =====
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ cat > Makefile <<'EOF'
.PHONY: check build clean

check:
    ./check.sh
```

图 15: q16 运行截图 24


```
===== 2 passed in 0.01s =====
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ cat > Makefile <<'EOF'
.PHONY: check build clean

check:
    ./check.sh

build:
    python3 -m build

clean:
    rm -rf build dist
EOF
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ make check
./check.sh
=== ruff format check ===
4 files already formatted
=== ruff lint check ===
All checks passed!
=== pytest run ===
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/lin/sdt-course/week4/q16
configfile: pyproject.toml
testpaths: .
collected 2 items

test_greet.py .. [100%]

===== 2 passed in 0.01s =====
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ make build
python3 -m build
* Creating isolated environment: venv+pip...
* Installing packages in isolated environment:
  - setuptools>=68
* Getting build dependencies for sdist...
running egg_info
writing src/greetlab_2026000000.egg-info/PKG-INFO
writing dependency_links to src/greetlab_2026000000.egg-info/dependency_links.txt
writing entry points to src/greetlab_2026000000.egg-info/entry_points.txt
writing top-level names to src/greetlab_2026000000.egg-info/top_level.txt
reading manifest file 'src/greetlab_2026000000.egg-info/SOURCES.txt'
```

图 16: q16 运行截图 5

```
running bdist_wheel
running build
running build_py
creating build/lib/greetlab
copying src/greetlab/cli.py -> build/lib/greetlab
copying src/greetlab/__init__.py -> build/lib/greetlab
running egg_info
writing src/greetlab_2026000000.egg-info/PKG-INFO
writing dependency_links to src/greetlab_2026000000.egg-info/dependency_links.txt
writing entry points to src/greetlab_2026000000.egg-info/entry_points.txt
writing top-level names to src/greetlab_2026000000.egg-info/top_level.txt
reading manifest file 'src/greetlab_2026000000.egg-info/SOURCES.txt'
writing manifest file 'src/greetlab_2026000000.egg-info/SOURCES.txt'
installing to build/bdist.linux-x86_64/wheel
running install
running install_lib
creating build/bdist.linux-x86_64/wheel
creating build/bdist.linux-x86_64/wheel/greetlab
copying build/lib/greetlab/cli.py -> build/bdist.linux-x86_64/wheel/./greetlab
copying build/lib/greetlab/__init__.py -> build/bdist.linux-x86_64/wheel/./greetlab
running install_egg_info
Copying src/greetlab_2026000000.egg-info to build/bdist.linux-x86_64/wheel/./greetlab_2026000000-0.1.0-py3.12.egg-info
running install_scripts
creating build/bdist.linux-x86_64/wheel/greetlab_2026000000-0.1.0.dist-info/WHEEL
creating '/home/lin/sdt-course/week4/q16/dist/.tmp-k662yf67/greetlab_2026000000-0.1.0-py3-none-any.whl' and adding 'build/bdist.linux-x86_64/wheel' to it
adding 'greetlab/__init__.py'
adding 'greetlab/cli.py'
adding 'greetlab_2026000000-0.1.0.dist-info/METADATA'
adding 'greetlab_2026000000-0.1.0.dist-info/WHEEL'
adding 'greetlab_2026000000-0.1.0.dist-info/entry_points.txt'
adding 'greetlab_2026000000-0.1.0.dist-info/top_level.txt'
adding 'greetlab_2026000000-0.1.0.dist-info/RECORD'
removing build/bdist.linux-x86_64/wheel
Successfully built greetlab_2026000000-0.1.0.tar.gz and greetlab_2026000000-0.1.0-py3-none-any.whl
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ sha256sum dist/*.whl
2c33f7032736c3d85dc322b59461b21545983e9d70d37f694e3e844c4b916b73 dist/greetlab_2026000000-0.1.0-py3-none-any.whl
(q16-venv) lin@lin-VirtualBox:~/sdt-course/week4/q16$ deactivate
lin@lin-VirtualBox:~/sdt-course/week4/q16$ rm -rf q16-venv build dist __pycache__ .pytest_cache src/greetlab/__pycache__
_ src/*.egg-info
lin@lin-VirtualBox:~/sdt-course/week4/q16$ cd ..
```

图 17: q16 运行截图 6

格式校验直接失败，脚本提前终止，`pytest` 没有运行，看不到业务逻辑带来的测试失败。

解决：执行 `ruff format .` 修复文件格式，保证格式门禁通过，让 `pytest` 可以执行，观察逻辑缺陷造成的测试断言失败。

- 问题 3: `Makefile` 伪目标 `check/build/clean` 未加入 `.PHONY` 声明；`build` 目标没有调用 `python3 -m build` 生成 `wheel` 包。

解决：添加 `.PHONY: check build clean`；`build` 配方调用 `build` 模块生成 `wheel`，`clean` 删除 `build`、`dist` 打包产物。

- 问题 4: 生成 `wheel` 后没有计算 `SHA-256` 哈希，缺少题目要求的完整性校验信息。

解决：使用 `sha256sum dist/*.whl` 获取 `wheel` 文件的 `sha256` 摘要。

3 课后练习

实例 1: 虚拟环境环境变量对比题目: 使用 `printenv` 将当前环境变量保存到 `before.txt`；创建 `Python` 虚拟环境并激活，再执行 `printenv` 保存到 `after.txt`；使用 `diff before.txt after.txt` 对比两份文件。观察环境发生了哪些变化？为什么 `Shell` 会优先使用虚拟环境内的程序？（提示：重点对比激活前后的 `$PATH` 变量）。执行 `which deactivate`，分析 `deactivate` 这个 `bash` 函数的工作原理。命令：

```
printenv > before.txt
python3 -m venv myvenv
source myvenv/bin/activate
printenv > after.txt
diff before.txt after.txt
which deactivate
```

输出：得到两份环境变量文件，`diff` 输出差异，观察 `PATH` 变量头部新增虚拟环境 `bin` 目录；`which deactivate` 无普通可执行文件输出，确认 `deactivate` 为 `shell` 内置函数，并非磁盘上独立程序。

实例 2: `grep` 正则查找代码中 `subprocess.Popen(..., shell=True)` 题目：编写正则表达式，使用 `grep` 命令行工具，在 `Python` 代码文件中查找 `subprocess.Popen` 调用且参数包含 `shell=True` 的代码行。命令：

```
grep -nE 'subprocess\.Popen\(.*shell=True\)\' *.py
```

输出：匹配到对应代码行并显示行号；若无匹配内容则无输出。示例命中输出：7: `proc = subprocess.Popen(["ls -l /tmp"], shell=True)`。

实例 3: 正则从 JSON 字符串捕获 name 字段值题目: 给定 JSON 字符串形如 {"name": "Alyssa P. Hacker", "college": "MIT"}, 编写正则表达式捕获 name 对应的字段值。命令:

```
echo '{"name": "Alyssa P. Hacker", "college": "MIT"}' | grep -oP '"name":\K[^"]+' 
```

输出:

```
Alyssa P. Hacker
```

实例 4: jq 过滤 JSON 数组, 筛选数值大于阈值的元素题目: 给定 JSON 数组, 筛选 downloads > 200 的对象, 仅输出对象的 name 字段。命令:

```
echo ' [{"name": "alpha", "downloads": 120}, {"name": "beta", "downloads": 900}, {"name": "gamma", "downloads": 450}] ' | jq -r 'map(select(.downloads>200))[]|.name'
```

输出:

```
beta
gamma
```

实例 5: find+xargs 递归统计项目下 py 文件总代码行数题目: 递归查找当前目录全部.py 文件, 统计全部 Python 文件的总行数; 使用 print0 与 xargs-0 兼容文件名含空格场景。命令:

```
find . -name "*.py" -type f -print0 | xargs -0 cat | wc -l
```

输出: 输出一个整数, 代表所有 py 文件合并后的总行数。

实例 6: sed 正则捕获组, 注释调试 print 语句题目: 使用 sed 捕获组, 将形如 print("debug: x = 123") 调试打印语句注释。命令:

```
echo 'print("debug: x=123")' | sed -E 's/^(print\(.*\))/#\1/'
```

输出:

```
# print("debug: x = 123")
```

实例 7: Make 伪目标演示, 观察.PHONY 作用题目: 创建名为 clean 普通文件, 编写 Makefile 不添加.PHONY, 执行 make clean; 再增加.PHONY: clean 再次执行, 对比两次行为差异。命令:

```
touch clean
cat > Makefile <<'EOF'
clean:
rm -f *.tmp
EOF
make clean
```

```
cat > Makefile <<'EOF'
.PHONY: clean
clean:
rm -f *.tmp
EOF
make clean
```

输出：无 PHONY 时提示 `make: Nothing to be done for 'clean';` 添加 PHONY 后会执行 `rm` 配方。

实例 8: `git log` 正则筛选 `feat` 类型提交记录题目：筛选 `git` 提交历史中提交信息以 `feat:` 开头的记录，简短输出哈希与提交说明。命令：

```
git log --grep='^feat:' --oneline
```

输出：打印匹配的 `commit` 短哈希与提交说明，例如：

```
a32bc1f feat: 完成第四周q13 建立可执行的本地质量门禁实验
67ad211 feat: 完成第四周q14 Make构建系统实验
```

实例 9: `ruff` 命令行检测未使用导入题目：使用 `ruff` 代码质量工具，检测 Python 源码中未使用的 `import`（F401 规则）。命令：

```
ruff check --select F401 src/
```

输出：命中未使用导入时输出文件路径、行号与 F401 告警；无问题则无告警输出。示例：`src/greetlab/cli.py:2:1: F401 `sys' imported but unused.`

实例 10: 管道统计模拟日志出现频率最高的 `ERROR` 关键词题目：给定模拟日志文本，提取 `ERROR` 后的关键词，统计出现频次，输出出现次数最高的 5 个关键词。命令：

```
cat <<'EOF' | grep -oP 'ERROR: \K\w+' | sort | uniq -c | sort -nr |
head -n5
ERROR: network
ERROR: disk
ERROR: network
ERROR: auth
ERROR: network
ERROR: disk
EOF
```

输出：

```
3 network
2 disk
1 auth
```

实例 11: `jq` 对 JSON 数组字段求和题目：模拟接口返回 JSON 数组，计算数组内所有 `downloads` 字段数值总和。命令：

```
echo ' [{"name":"a","downloads":10},{ "name":"b","downloads":20},{ "name":"c","downloads":30}] ' | jq '[.[] .downloads] | add'
```

输出:

```
60
```

4 解题感悟

第四周实验学习了代码质量门禁、**Make** 构建系统、命令行 **JSON** 处理、综合交付测试。本地质量门禁可以将格式化、静态检查、单元测试整合到一条脚本，在提交代码前快速发现问题，提前规避低级错误。**Make** 工具依靠文件时间戳判断是否重新构建，需要准确维护依赖关系，伪目标**PHONY** 是容易踩坑的关键点。**curl+jq** 可以直接在 **shell** 完成 **JSON** 数据过滤、排序、报告生成，不需要编写 **Python** 脚本。综合交付测试需要完成缺陷复现、定位修复、打包、完整性哈希校验的完整流程。本次实验继续坚持一题一 **commit**, **git** 历史清晰，方便回溯每一步改动。**shell** 正则、管道工具在处理文本、日志、**JSON** 数据时非常高效，适合自动化小脚本开发。

5 代码仓库提交记录

GitHub 仓库链接: <https://github.com/imnotab/sdtb/tree/main>



图 18: 项目 commit 历史记录